Basic Python Programming

In [1]:
```
import sys
sys.version
```

Out[1]:
```
'3.11.7 | packaged by Anaconda, Inc. | (main, Dec 15 2023, 18:05:47) [MSC v.1916 64 bit (AMD6
4)]'
```

Work With Numbers

In [25]:
```
3
```

Out[25]:
```
3
```

In [27]:
```
2
```

Out[27]:
```
2
```

In [29]:
```
3+2
```

Out[29]:
```
5
```

In [31]:
```
3-2
```

Out[31]:
```
1
```

In [33]:
```
3*2
```

Out[33]:
```
6
```

In [35]:
```
3**2
```

Out[35]:
```
9
```

In [37]:
```
3**2
```

Out[37]:
```
9
```

In [39]:
```
10/5
```

Out[39]:
```
2.0
```

In [41]:
```
10//5
```

Out[41]:
```
2
```

In [ ]:
```
# wotk with String
```

In [43]:
```
helloworld
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[43], line 1
----> 1 helloworld

NameError: name 'helloworld' is not defined
```

In [45]:
```
'helloworld'
```

Out[45]:
```
'helloworld'
```

In [47]:
```
"helloworld"
```

Out[47]:
```
'helloworld'
```

In [ ]:
```

```

In [49]:
```
'''helloworld'''
```

Out[49]:
```
'helloworld'
```

Variable = object

In [51]:
```
v = 5 # v - variable and 5 for value
v
```

Out[51]:
```
5
```

In [53]:
```
type(v)
```

Out[53]:
```
int
```

In [57]:
```
v1= 'hello'
v1
```

Out[57]:
```
'hello'
```

In [ ]:
```
v1
```

26 th

In [59]:
```
a = 5.5
type(a)
```

Out[59]:
```
float
```

In [61]:
```
import sys
sys.version
```

Out[61]:
```
'3.11.7 | packaged by Anaconda, Inc. | (main, Dec 15 2023, 18:05:47) [MSC v.1916 64 bit (AMD6
4)]'
```

In [63]:
```
nit = 15 # Python is Case-sensitive
NIT
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[63], line 2
      1 nit = 15
----> 2 NIT

NameError: name 'NIT' is not defined
```

In [65]:
```
nit
```

Out[65]:
```
15
```

In [67]:
```
1a = 67 # Rule 2. Python Veriable never starts with integer but ends with integer
1a
```

```
  Cell In[67], line 1
    1a = 67
    ^
SyntaxError: invalid decimal literal
```

In [69]:
```
a1 =67 # Veriable ends with integer
a1
```

Out[69]:
```
67
```

In [71]:
```
nit$ = 89 #Rule 3. Python Never use special symbol except _(Underscore)
nit$
```

```
  Cell In[71], line 1
    nit$ = 89
       ^
SyntaxError: invalid syntax
```

In [92]:
```
x_train, x_test, y_train, y_test = 80, 20, 70, 30 # here if you decare multiple veriable but
```

In [102…
```
x_train
x_test
y_train
y_test
```

Out[102…
```
30
```

In [106…
```
print(x_train)
print(x_test)
print(y_train)
print(y_test)
```

```
80
20
70
30
```

In [110…
```
import keyword
keyword.kwlist
```

Out[110…

```
['False',
 'None',
 'True',
 'and',
 'as',
 'assert',
 'async',
 'await',
 'break',
 'class',
 'continue',
 'def',
 'del',
 'elif',
 'else',
 'except',
 'finally',
 'for',
 'from',
 'global',
 'if',
 'import',
 'in',
 'is',
 'lambda',
 'nonlocal',
 'not',
 'or',
 'pass',
 'raise',
 'return',
 'try',
 'while',
 'with',
 'yield']
```

In [112…

```
if = 90 # never use Predefine keyword
if
```

```
  Cell In[112], line 1
    if = 90
       ^
SyntaxError: invalid syntax
```

In [116…

```
a10 = 78
a9 = 89
```

In [118…

```
print(a10)
print(a9)
```

```
78
89
```

In [120…

```
del a10 # here if you del funcation then it will delete value from memory. for that reaseon y
```

In [122…

```
a10
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[122], line 1
----> 1 a10

NameError: name 'a10' is not defined
```

In [130…

```
for = 90
```

```
  Cell In[130], line 1
    for = 90
        ^
SyntaxError: invalid syntax
```

In [139…
```
For = 90
For
```

Out[139…
```
90
```

In [143…
```
a = True
a
```

Out[143…
```
True
```

In [145…
```
b = 'True'
b
```

Out[145…
```
'True'
```

In [147…
```
pi = 3.17 # pi value never prefix
pi
```

Out[147…
```
3.17
```

In [149…
```
pi = 3.20
pi
```

Out[149…
```
3.2
```

In [151…
```
aaaaaaaaaaaaaaaaaaaaaaa = 90 #never declare variable such way
aaaaaaaaaaaaaaaaaaaaaaa
```

Out[151…
```
90
```

In [ ]:
```
```